

Morphological Operations

Image Processing & Analysis for Life Scientists

Olivier Burri, Romain Guiet & Arne Seitz

Morphological Operations

1. Open *Leaf (36 k)*
File→Open Samples→Leaf (36 k)
2. Convert the image to 8-bit.
3. Crop the image.
Only the leaf shall be visible afterwards.

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1. Open *Leaf (36 k)*
File→Open Samples→Leaf (36 k)
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3. Make a selection , so only the leaf shall be visible afterwards.
4. Image→Crop the image.



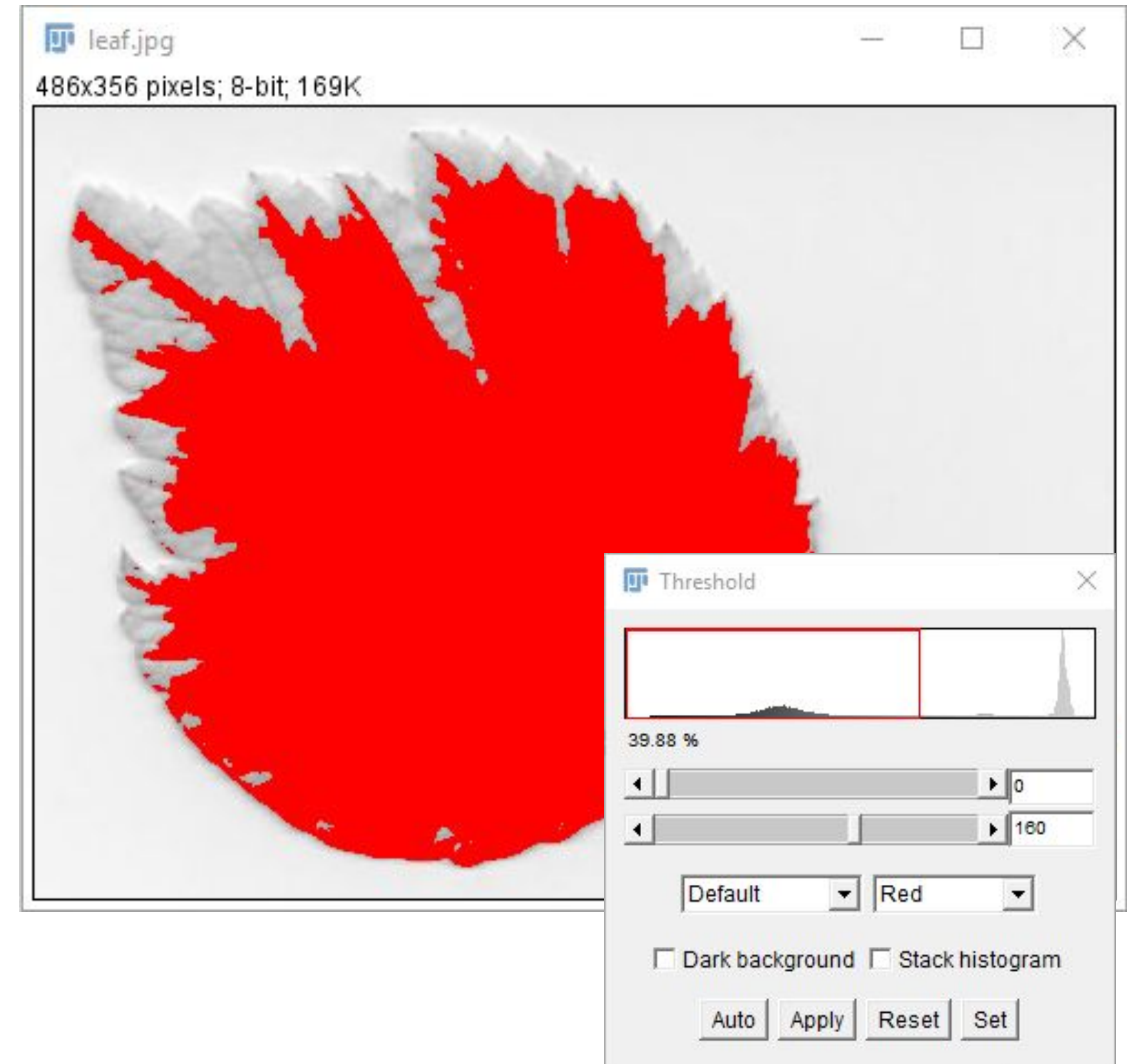
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4. Image→Crop the image.
- 5.** Duplicate the image twice.
- 6.** Select one of the duplicated images.
- 7.** Use simple thresholding in order to select the inner part of the leaf.



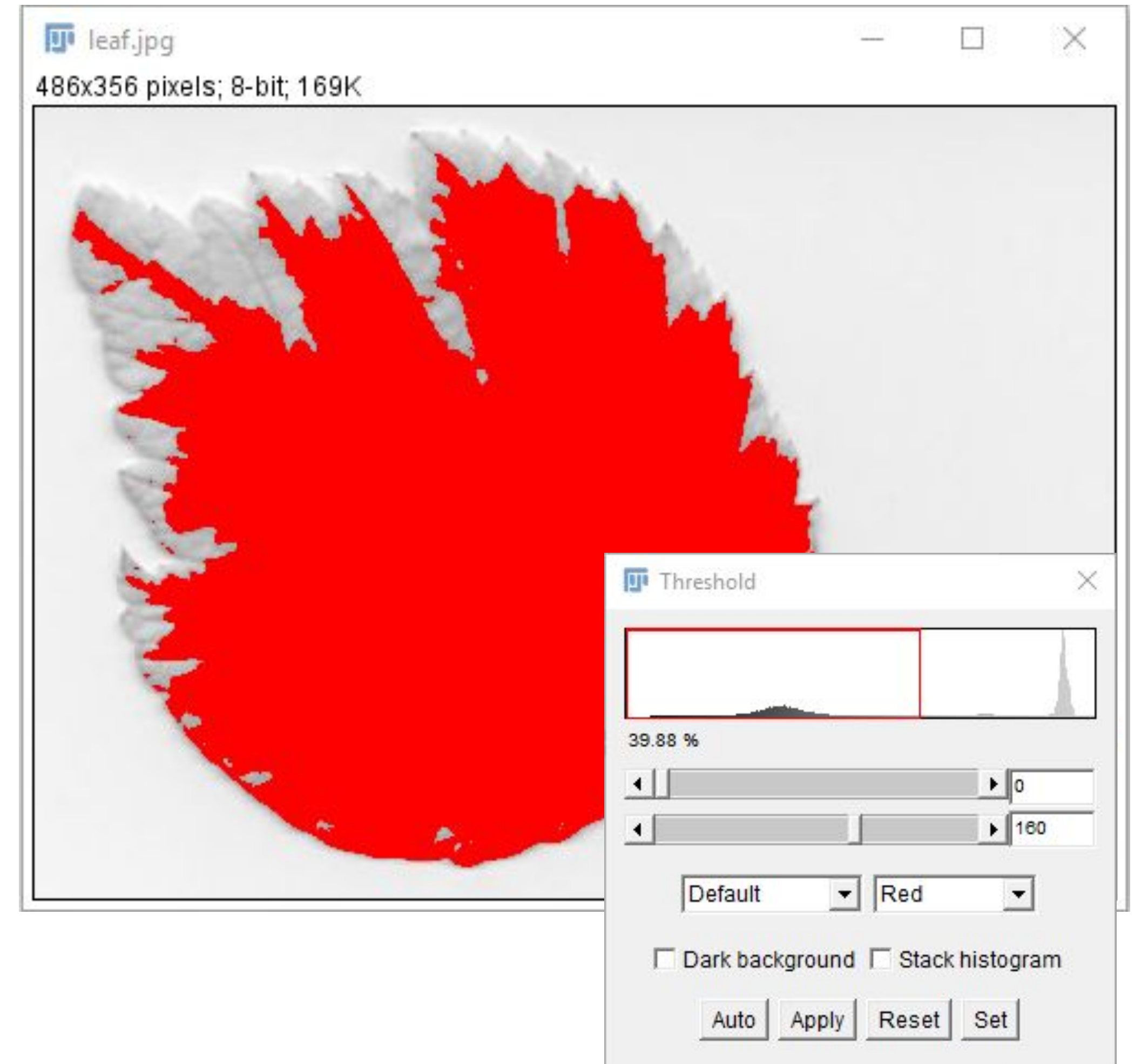
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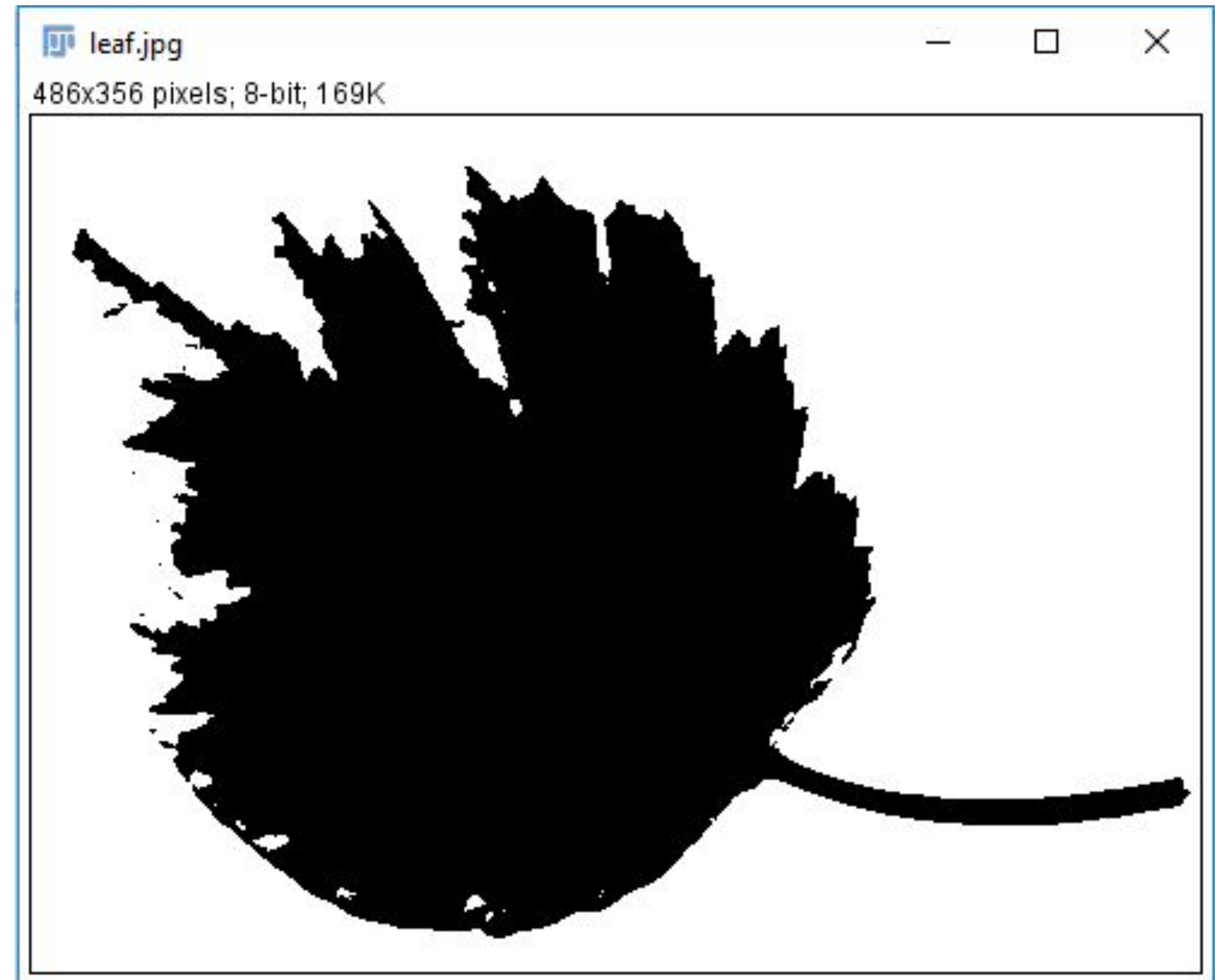
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- 7. Convert the image to a binary image.**



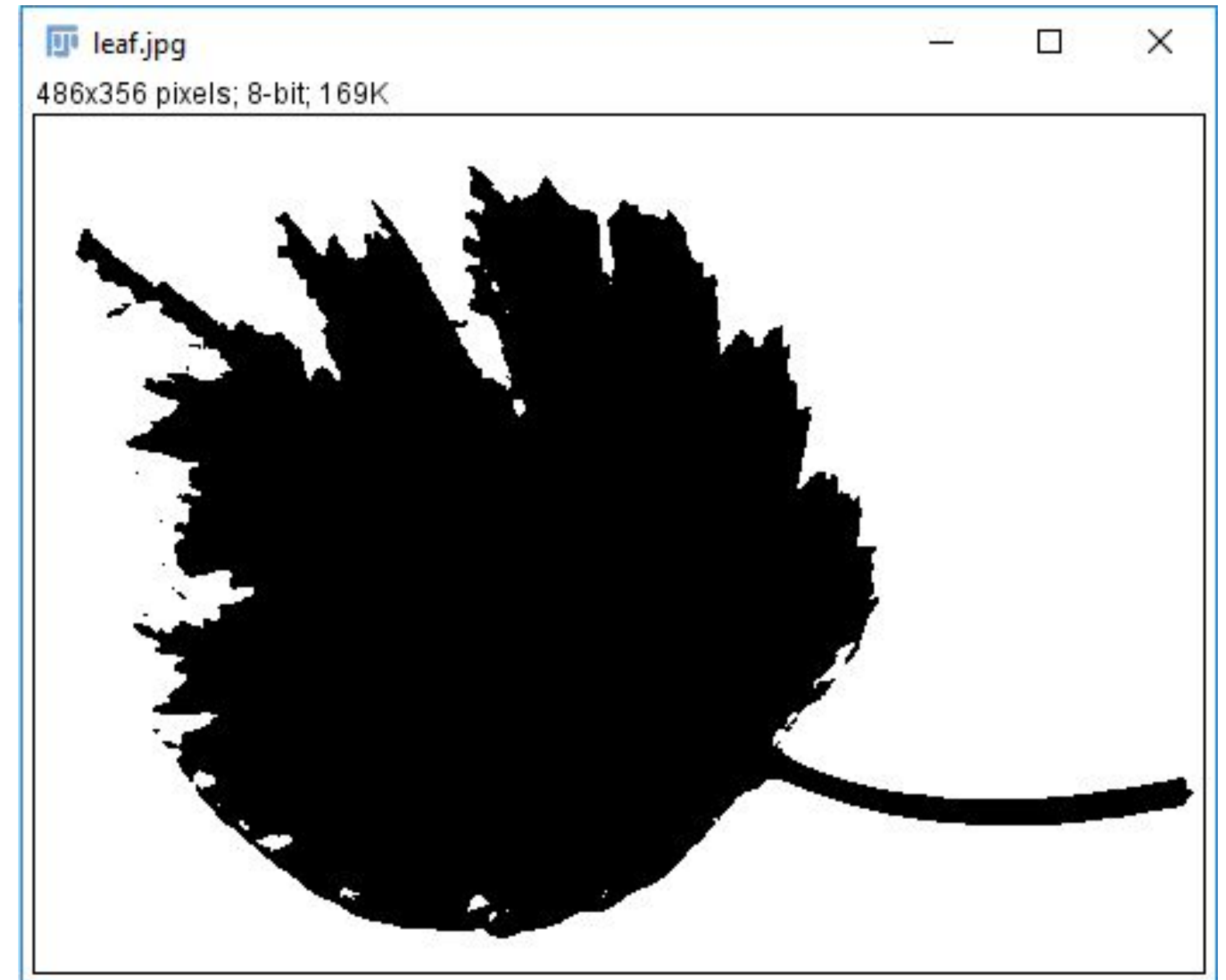
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- 7. Convert the image to a binary image.**
Process→Binary→Make Binary



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7. Convert the image to a binary image.
Process→Binary→Make Binary
- 8. Apply Outline and Dilate to the image
(one after the other)**



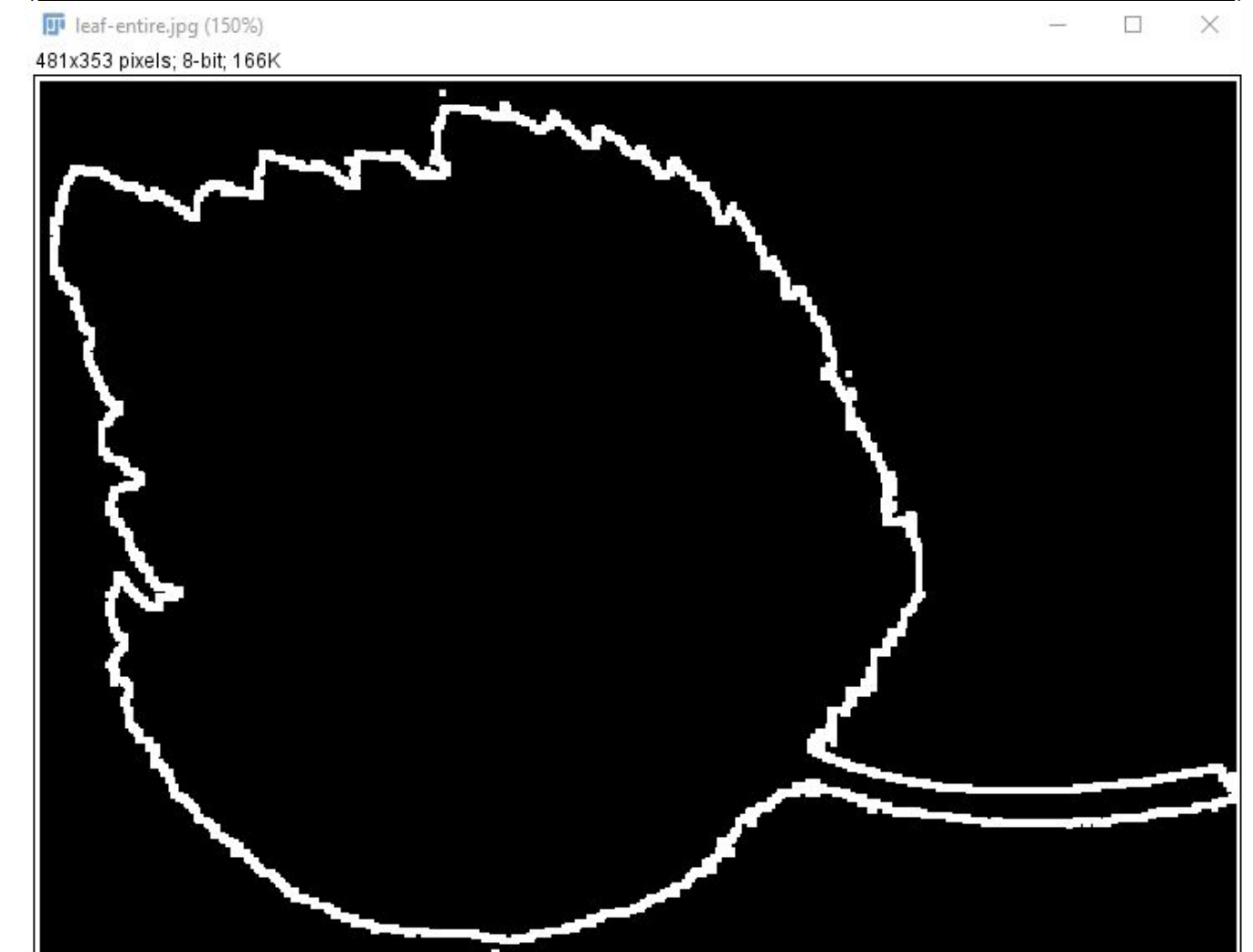
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Process→Binary→Make Binary
- 8. Apply Outline and Dilate to the image**
Process→Binary→Outline
Process→Binary→Dilate



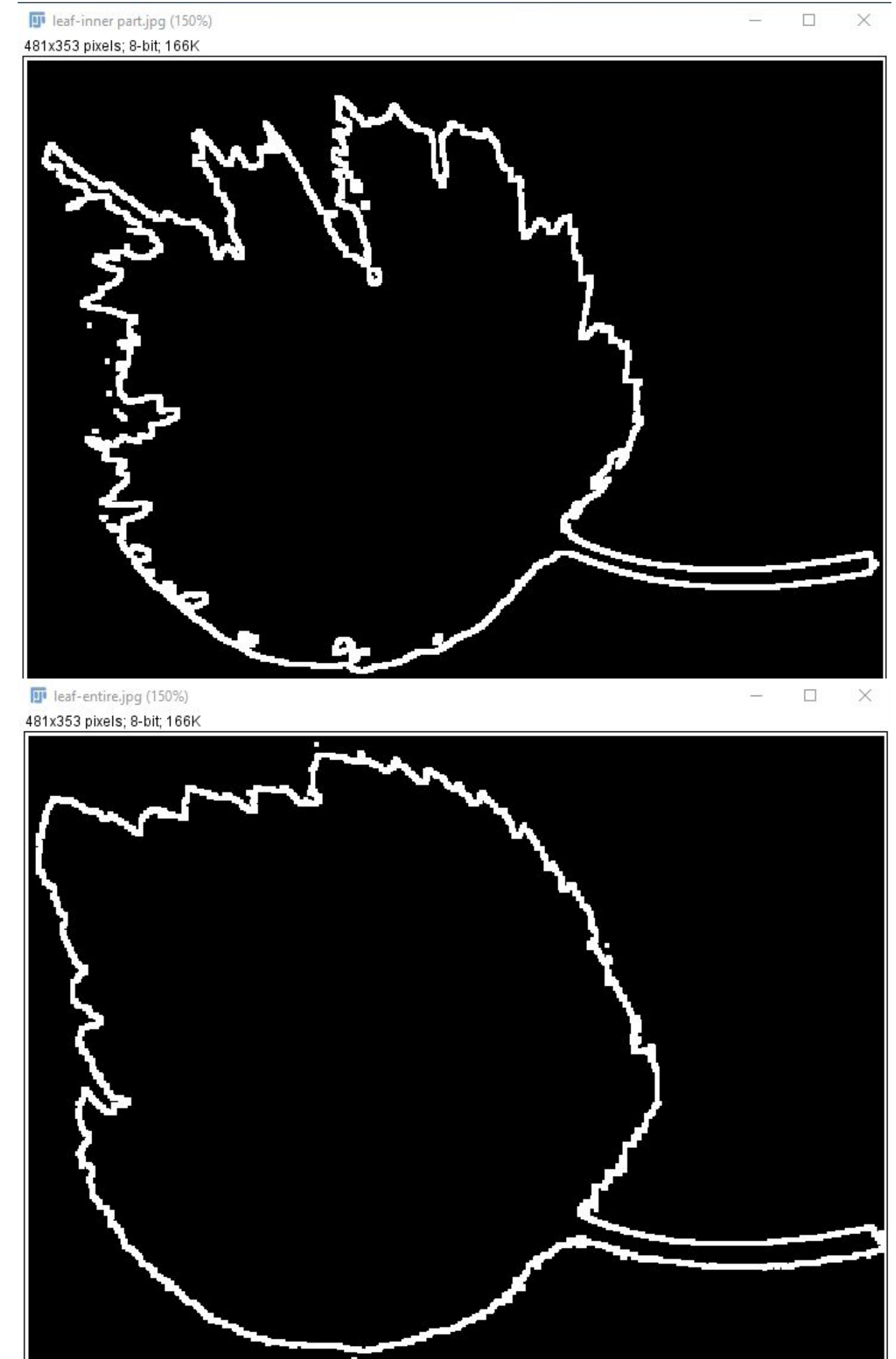
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8. Apply Outline and Dilate to the image
Process→Binary→Outline
Process→Binary→Dilate
9. Rename the image.
10. Repeat 5.-10. to outline the entire leaf.



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8. Apply Outline and Dilate to the image
Process→Binary→Outline
Process→Binary→Dilate
9. Rename the image.
10. Repeat 5.-10. to outline the entire leaf.
11. Invert the original image.
12. Make an overlay of the inverted original image and the created outlines.



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8. Apply Outline and Dilate to the image
Process→Binary→Outline
Process→Binary→Dilate
9. Rename the image.
10. Repeat 5.-10. to outline the entire leaf.
11. Invert the original image.
12. Make an overlay of the inverted original image and the created outlines.
Image→Color→Merge Channels...

